

Business Finance Pro

Version 1.0.0 — Macro-Free Edition

Complete Technical & User Documentation

*A self-contained financial operating system for small business owners,
built entirely in Microsoft Excel — no macros, no add-ins, no external services.*

1. Executive Summary

Purpose

Business Finance Pro replaces the mental overhead of formal accounting with a plain-language, five-minute weekly ritual: log what happened, glance at your health, act on what needs attention. It is a finished product built entirely inside Excel, not a template the customer configures themselves.

Target Users

Freelancers, shop owners, and small service businesses with little or no formal accounting training — owners who currently avoid looking at their numbers because every existing tool assumes either an accountant's vocabulary or hours per week they don't have.

Key Capabilities

- A live Home dashboard summarizing business health, monthly cash flow, urgent action items, and a plain-language weekly narrative
- Guided entry screens for transactions, bills, invoices, and savings/revenue goals
- Three independent rule-based calculation engines plus a shared metrics core — zero AI, zero VBA, formulas only
- A fully validated, protected data layer that prevents accidental corruption while remaining transparent and auditable

Design Philosophy

Every design decision was tested against one question: would a non-accountant understand this without help? The workbook avoids accounting terminology, exposes only plain-language design tokens (not raw numbers or jargon) to the presentation layer, and treats calm, non-judgmental tone as a hard requirement. Structurally, it separates data, calculation, and presentation into three strict layers — the same pattern a real SaaS product would use, adapted to what native Excel can do without macros.

2. Feature List

Module	Status	Notes
Home Dashboard	Complete	Business Health, Money In/Out, Needs Attention, Monday Briefing
Settings	Complete	11 editable fields, 4 read-only Product Information fields
Quick Add	Complete	Guided transaction entry with staging workflow

Bills	Complete	Bill list + add-a-bill workflow; feeds the Attention engine
Clients	Complete	Invoice list + add-an-invoice workflow; feeds the Attention engine
Goals	Complete	Goal list with live progress + add-a-goal workflow; feeds Insights
Hidden Data Layer	Complete	9 protected Excel Tables, fully validated
Calculation Engines	Complete	4 independent, modular rule engines
Insights	Complete	Top-3 ranked narrative insights, rule-based
Business Health	Complete	6-state weighted scoring engine, design-token outputs
Needs Attention	Complete	Up to 5 ranked, prioritized action items
Navigation	Complete	Fully wired hyperlinks across all 6 visible sheets
Data Validation	Complete	46 rules across 16 reusable dropdown lists
Protection	Complete	Every hidden sheet locked; visible sheets locked except inputs
Named Ranges	Complete	106 named ranges, the sole cross-layer interface

3. Workbook Architecture

Sheet Inventory (20 total)

Visible (6): Home, Settings, Quick Add, Bills, Clients, Goals

Hidden (14): _data_Transactions, _data_Categories, _data_Vendors, _data_Bills, _data_Customers, _data_Invoices, _data_Goals, _data_BusinessProfile, _data_Settings, _calc_Core, _calc_Health, _calc_Attention, _calc_Insights, _lists_Validation

Data Flow

User input (visible sheets) → staging rows → manual commit → hidden data tables → _calc_Core (aggregates) → _calc_Health / _calc_Attention / _calc_Insights → named ranges → Home Dashboard.

Calculation Flow

Every visible sheet consumes calculation results exclusively through named ranges — never through direct references to a hidden table or another calculation sheet, with one documented exception: Bills, Clients, and Goals list cards read their own tables directly. No calculation engine references a presentation sheet. This one-directional flow keeps the product debuggable and prevents circular logic.

4. Data Model

Table	Purpose
tbl_Transactions	The atomic record of every dollar moving in or out of the business
tbl_Categories	Plain-language income/expense classification, pre-loaded with 12 defaults
tbl_Vendors	Anyone the business pays
tbl_Customers	Anyone who pays the business
tbl_Bills	Money owed to vendors — one-time or recurring
tbl_Invoices	Money owed by customers
tbl_Goals	Savings, revenue, or debt-payoff targets with live progress tracking
tbl_BusinessProfile	Singleton record of business identity
tbl_Settings	Singleton record of user preferences and product metadata

5. Calculation Engine Overview

`_calc_Core`

The foundation every other engine builds on. Computes seven month-scoped metrics (Money In, Money Out, Net Cash Flow, Transaction Count, Average Transaction, Current Month, Previous Month) directly from `tbl_Transactions` using structured references only.

`_calc_Health`

A four-section, weighted-component scoring engine (Inputs → Intermediate → Threshold Evaluation → Output) producing a plain-language business health state. Restricted by design to consuming only `_calc_Core`. Currently scores on two real components (Cash Flow, Activity); three components (Cash Runway, Bills, Invoices) exist as architecturally-reserved placeholders.

`_calc_Attention`

A rule-based engine evaluating up to 7 candidate conditions (5 active, 2 reserved), surfacing up to 5 prioritized, ranked action items. Consumes `_calc_Core` plus direct structured references to `tbl_Bills` and `tbl_Invoices`.

`_calc_Insights`

A rule-based narrative engine producing the Top 3 "Monday Briefing" insights shown on Home, ranked from 6 candidate rules (5 active). Combines `_calc_Core` aggregates with category/vendor spend breakdowns and goal progress to write plain-English observations.

6. User Guide

Configure Settings

Open Settings, fill in Business Information, Appearance, and Preferences. The four Product Information fields are system-set and read-only.

Add a Transaction

Go to Quick Add, fill in the seven fields under "What happened?" A staging row builds itself. Copy it and paste it (as values) into the first empty row of your Transactions table.

Add a Bill

Same pattern on the Bills screen: fill in "Add a bill," copy the staging row into your Bills table.

Add a Client / Invoice

Same pattern on the Clients screen.

Add a Goal

Same pattern on the Goals screen. New goals start at \$0 progress and Active status.

Read the Dashboard

Home shows Business Health, this month's Money In/Out, up to five Needs Attention items, and a weekly narrative under "This Week."

Understand Business Health

A single plain-language state tells you whether income is comfortably ahead of spending this month. It is intentionally not a number.

Understand Insights

The "This Week" narrative combines up to three automatically-selected observations — biggest expense, top vendor, income, activity, and goal progress.

Understand Needs Attention

Up to five ranked items requiring action: no income logged, spending exceeding income, overdue bills, overdue invoices.

7. Known V1 Limitations

- **Macro-free staging workflow.** Every add screen requires a manual copy-and-paste-as-values step to permanently save an entry — a structural consequence of Excel formulas being unable to write and reset themselves, not fixable without VBA.
- **Static currency formatting.** Currency is stored and correctly editable in Settings, but every dollar amount uses a hardcoded \$ symbol. Changing Currency does not change any displayed symbol.
- **Placeholder score components.** _calc_Health's Cash Runway, Bills, and Invoice score components are architecturally present but return zero.
- **Placeholder Attention rule.** An "Upcoming loan payment" rule exists as a labeled placeholder; no Loans table exists in V1.
- **Single-goal insight visibility.** Only the highest-priority Active goal is ever mentioned in Insights.
- **List-card architectural exception.** Bills, Clients, and Goals display their lists via direct table references rather than the named-ranges-only pattern used on Home.
- **Three Settings fields have no UI.** InsightsEnabled, CelebrationAnimationsEnabled, and DefaultInvoiceDueDays are real, working fields with no way for a user to change them from any screen.

8. Technical Statistics

Metric	Count
Workbook version	1.0.0 (Macro-Free Edition)
Total sheets	20
Visible sheets	6
Hidden sheets	14
Excel Tables	9
Named ranges	106
Formulas	425
Validation lists	16
Data validation rules	46
Conditional formatting rules	31
Calculation engines	4

9. Version History

The workbook was built in strict dependency order across roughly 20 development sprints, organized into five phases:

- **Product definition** — PRD, personas, UX principles, design system, and data model, all locked before construction began.
- **Foundation** — workbook specification, Home and Settings construction, the hidden data layer, and the binding layer connecting them via named ranges.
- **Calculation engines** — `_calc_Core`, `_calc_Health` (later refined to a weighted-component architecture), `_calc_Attention`, and `_calc_Insights`, each built and tested independently.
- **Remaining modules** — Quick Add, Bills, Clients, and Goals, each activating a previously-placeholder calculation rule as its underlying data became available.
- **Release preparation** — a full RC1 structural audit and an independent adversarial RC2 QA review, both of which found and fixed genuine bugs before this documentation was produced.

10. Future Roadmap (V2 — Not Implemented)

The following require VBA, Office Scripts, or a technology beyond native Excel formulas, and are explicitly out of scope for this release:

- One-click data entry, eliminating the manual staging/copy-paste step
- Dynamic currency symbols driven by the Currency setting
- Full Business Health scoring with Cash Runway, Bills, and Invoice components activated
- Multi-goal insight support
- A Loans module (`tbl_Loans` and associated engine rules)
- Recurring bill/transaction automation
- UI for the three currently-orphaned Settings preference fields
- A companion mobile capture tool that syncs back into this workbook

11. Final Production Assessment

Status: Release Candidate, audited and QA-verified.

The core financial logic, calculation engines, data layer, and navigation are complete, tested against real recalculation across two independent audit cycles, and free of known bugs at time of publication. The product's central interaction — saving a transaction, bill, invoice, or goal — requires a manual copy-paste step that is a deliberate, documented architectural characteristic of the macro-free approach, not an oversight.

This workbook is production-ready as a macro-free V1 release. A future release introducing VBA or Office Scripts could remove the manual staging step entirely without altering anything else documented here.